The threshold $a = \mathrm{e}$ for $(x-1)a^x + 1 \ge x^a$

monotonicity, a double contact at $x = 0$ and $x = 1$, and two one-sided proofs

Problem. Let $a > 1$ and

$$f(x) = (x-1)a^x + 1.$$

(i) For $a > \mathrm{e}$, determine the intervals of monotonicity of f.

(ii) Find all $a > 1$ for which

$$(x-1)a^x + 1 \ge x^a \qquad \text{for every } x \ge 0.$$

$$\boxed{\text{(i)}\quad f\downarrow \text{ on } (-\infty, x_0), \quad f\uparrow \text{ on } (x_0, \infty), \quad x_0 = 1 - \frac{1}{\log a}; \qquad \text{(ii)}\quad 1 < a \le \mathrm{e}.}$$

Overview. Throughout put

$$\lambda = \log a, \qquad a = \mathrm{e}^\lambda,$$

which turns the threshold $a = \mathrm{e}$ into $\lambda = 1$. Part (i) is a one-line derivative computation: the exponential factor is positive, so the sign of f' is the sign of a linear function.

Part (ii) is governed by the difference

$$F(x) = 1 + (x-1)\mathrm{e}^{\lambda x} - x^{\mathrm{e}^\lambda},$$

which vanishes at *two* points, $x = 0$ and $x = 1$. The necessity of $a \le \mathrm{e}$ is read off the slope at $x = 0$, where $F(x) \sim (1-\lambda)x$. Sufficiency splits at the second contact point: on $(0,1)$ a logarithmic-derivative comparison gives $g(x) \ge x^a$ directly, and on $[1, \infty)$ one shows $F' \ge 0$, so F grows away from $F(1) = 0$. The point $x = 1$ is not an arbitrary cut: it is where the two curves touch a second time.

1. Part (i): monotonicity

Differentiating,

$$f'(x) = a^x + (x-1)a^x \log a = a^x\big(1 + (x-1)\log a\big). \tag{1}$$

Since $a^x > 0$ for all real x, the sign of f' is the sign of the affine factor $1 + (x-1)\log a$, which is increasing in x and vanishes at

$$x_0 = 1 - \frac{1}{\log a}. \tag{2}$$

If $a > \mathrm{e}$ then $\log a > 1$, so $0 < \frac{1}{\log a} < 1$ and

$$0 < x_0 < 1.$$

Hence

$$\boxed{f' < 0 \text{ on } (-\infty, x_0), \qquad f' > 0 \text{ on } (x_0, \infty), \qquad x_0 = 1 - \frac{1}{\log a}.}$$

Restricted to $x \ge 0$, which is the range relevant to part (ii), f decreases on $[0, x_0]$ and increases on $[x_0, \infty)$.

For $1 < a \le \mathrm{e}$ the same computation gives $x_0 \le 0$, so f is increasing on the whole of $[0, \infty)$. This fact reappears in Section 4 as the positivity of g'.

2. The two contact points

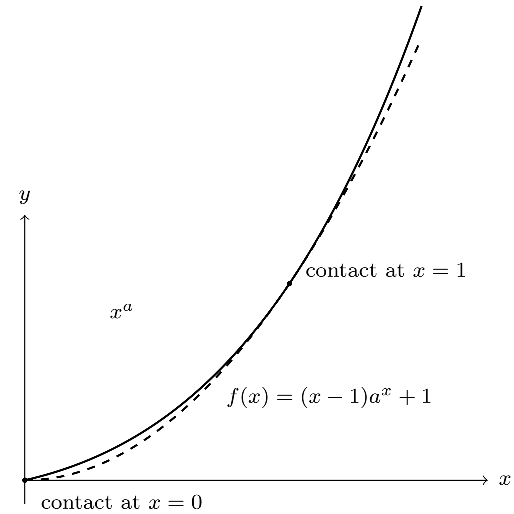
Set

$$F(x) = f(x) - x^a = 1 + (x-1)\mathrm{e}^{\lambda x} - x^{\mathrm{e}^{\lambda}}, \qquad x \geq 0.$$

Part (ii) asks for which $\lambda > 0$ we have $F \geq 0$ on $[0, \infty)$. Immediately

$$F(0) = 1 - 1 - 0 = 0, \qquad F(1) = 1 + 0 - 1 = 0, \tag{3}$$

using $a > 1$ for $0^a = 0$. So the graph of f meets the graph of x^a at both $x = 0$ and $x = 1$, and F must not dip below zero between or beyond them. Any such problem is decided by local behaviour at the contact points, and the softer of the two is $x = 0$.



$a = 2$: the solid curve stays above the dashed one

3. Necessity of $a \leq \mathrm{e}$

Expand near $x = 0^+$. Since $\mathrm{e}^{\lambda x} = 1 + \lambda x + O(x^2)$,

$$1 + (x-1)\mathrm{e}^{\lambda x} = 1 + (x-1)\big(1 + \lambda x + O(x^2)\big) = (1-\lambda)x + O(x^2). \tag{4}$$

Moreover $a > 1$ gives $x^a = o(x)$ as $x \to 0^+$. Hence

$$\frac{F(x)}{x} \longrightarrow 1 - \lambda, \qquad x \to 0^+. \tag{5}$$

If $\lambda > 1$, then $1 - \lambda < 0$, so $F(x) < 0$ for all sufficiently small $x > 0$ and the inequality fails. Therefore

$$\boxed{\lambda \leq 1, \qquad \text{i.e.} \qquad 1 < a \leq \mathrm{e}.} \tag{6}$$

At the borderline $\lambda = 1$ the linear term in (4) vanishes and

$$1 + (x-1)\mathrm{e}^{x} = \frac{x^2}{2} + \frac{x^3}{3} + \cdots,$$

while $x^a = x^{\mathrm{e}}$ with $\mathrm{e} > 2$, so $x^{\mathrm{e}} = o(x^2)$ and $F(x) \sim \frac{x^2}{2} > 0$. The threshold is thus decided by a comparison of *orders*: at $a = \mathrm{e}$ the quadratic term saves the inequality, and the exponent $\mathrm{e} > 2$ of the competing power is exactly what makes it lose.

4. Sufficiency on $0 < x < 1$

Assume $0 < \lambda \le 1$ and set

$$g(x) = 1 + (x-1)\mathrm{e}^{\lambda x} = 1 - (1-x)\mathrm{e}^{\lambda x},$$

so that $F = g - x^a$ and the claim is $g(x) \ge x^a$.

Positivity of g. By (1) with $f = g$,

$$g'(x) = \mathrm{e}^{\lambda x}\big(1 - \lambda(1-x)\big) \ge 0 \qquad \text{for } 0 \le x \le 1, \tag{7}$$

because $\lambda(1-x) \le \lambda \le 1$. Since $g(0) = 0$ and $g(1) = 1$, the function g increases from 0 to 1 on $[0,1]$; in particular $g > 0$ on $(0,1]$.

The comparison of logarithmic derivatives. We claim

$$\frac{x\,g'(x)}{g(x)} \le a \qquad \text{for } 0 < x < 1. \tag{8}$$

Granting (8), divide by $x > 0$ and integrate over $[x,1]$:

$$\int_x^1 \frac{g'(t)}{g(t)}\,\mathrm{d}t \le \int_x^1 \frac{a}{t}\,\mathrm{d}t.$$

The left side is $\log g(1) - \log g(x) = -\log g(x)$ since $g(1) = 1$; the right side is $-a\log x$. Hence $\log g(x) \ge a \log x$ and, exponentiating,

$$g(x) \ge x^a, \qquad 0 < x < 1. \tag{9}$$

This is the natural form of the argument: x^a is characterised by having logarithmic derivative exactly $\frac{a}{x}$, so (8) says that g grows no faster than x^a on the way *up* to the common value 1 at $x = 1$, and therefore lies above it on the way *down*.

Proof of (8). Since $x > 0$ and $g(x) > 0$, (8) is equivalent to

$$a\,g(x) - x\,g'(x) \ge 0.$$

Substituting $a = \mathrm{e}^{\lambda}$ and (7), and putting $y = 1 - x \in (0,1)$,

$$\begin{aligned} a\,g(x) - x\,g'(x) &= \mathrm{e}^{\lambda}\big(1 - y\,\mathrm{e}^{\lambda x}\big) - (1-y)\mathrm{e}^{\lambda x}\big(1 - \lambda y\big) \\ &= \mathrm{e}^{\lambda}\mathrm{e}^{-\lambda y}\Big[\mathrm{e}^{\lambda y} - \mathrm{e}^{\lambda} y - (1-y)(1-\lambda y)\Big] = \mathrm{e}^{\lambda x} H(y), \end{aligned}$$

where

$$H(y) := \mathrm{e}^{\lambda y} - \mathrm{e}^{\lambda} y - (1-y)(1-\lambda y). \tag{10}$$

It remains to prove $H \ge 0$ on $[0,1]$. Expanding both exponentials and cancelling,

$$H(y) = \lambda y(1-y) - \sum_{n\ge 2} \frac{\lambda^n\big(y - y^n\big)}{n!}. \tag{11}$$

For $0 \le y \le 1$,

$$y - y^n = y(1-y)\big(1 + y + \cdots + y^{n-2}\big) \le (n-1)\,y(1-y),$$

so

$$H(y) \ge y(1-y)\left[\lambda - \sum_{n\ge 2}\frac{(n-1)\lambda^n}{n!}\right]. \tag{12}$$

The series is elementary:

$$\sum_{n\ge 2}\frac{(n-1)\lambda^n}{n!} = \sum_{n\ge 2}\left[\frac{\lambda^n}{(n-1)!} - \frac{\lambda^n}{n!}\right] = \lambda\big(\mathrm{e}^{\lambda} - 1\big) - \big(\mathrm{e}^{\lambda} - 1 - \lambda\big) = (\lambda - 1)\mathrm{e}^{\lambda} + 1.$$

Therefore the bracket in (12) equals

$$\lambda - (\lambda - 1)\mathrm{e}^{\lambda} - 1 = (1 - \lambda)\big(\mathrm{e}^{\lambda} - 1\big) \geq 0 \qquad \text{for } 0 < \lambda \leq 1,$$

so $H \geq 0$, which proves (8) and hence (9):

$$\boxed{F(x) \geq 0 \qquad (0 < x < 1).}$$

Note where the hypothesis is used: the bracket is $(1 - \lambda)(\mathrm{e}^{\lambda} - 1)$, which changes sign exactly at $\lambda = 1$. The threshold of part (ii) is visible a second time, now in the interior estimate.

5. Sufficiency on $x \geq 1$

Here it is cleanest to show that F is increasing, so that it never returns below $F(1) = 0$. Differentiating and putting $u = x - 1 \geq 0$,

$$F'(x) = \mathrm{e}^{\lambda x}\big(1 + \lambda(x - 1)\big) - a\, x^{a-1} = \mathrm{e}^{\lambda}\Big[\mathrm{e}^{\lambda u}(1 + \lambda u) - (1 + u)^{\mathrm{e}^{\lambda} - 1}\Big].$$

Since $\mathrm{e}^{\lambda} > 0$, it suffices that

$$\mathrm{e}^{\lambda u}(1 + \lambda u) \geq (1 + u)^{\mathrm{e}^{\lambda} - 1},$$

and both sides are positive, so we may take logarithms. Define

$$R(u) = \lambda u + \log(1 + \lambda u) - \big(\mathrm{e}^{\lambda} - 1\big)\log(1 + u), \qquad R(0) = 0.$$

It suffices to prove $R \geq 0$ on $[0, \infty)$, and for that it suffices that $R' \geq 0$. Now

$$R'(u) = \lambda + \frac{\lambda}{1 + \lambda u} - \frac{\mathrm{e}^{\lambda} - 1}{1 + u} = \lambda\left(1 + \frac{1}{1 + \lambda u}\right) - \frac{\mathrm{e}^{\lambda} - 1}{1 + u}. \tag{13}$$

Two elementary estimates finish the argument.

Estimate A. For $u \geq 0$ and $0 < \lambda \leq 1$,

$$1 + \frac{1}{1 + \lambda u} \geq \frac{2}{1 + u},$$

because

$$1 + \frac{1}{1 + \lambda u} - \frac{2}{1 + u} = \frac{u\big(2 - \lambda + \lambda u\big)}{(1 + \lambda u)(1 + u)} \geq 0.$$

Estimate B. For $0 \leq \lambda \leq 1$,

$$\mathrm{e}^{\lambda} - 1 \leq 2\lambda.$$

Indeed $\psi(\lambda) = 1 + 2\lambda - \mathrm{e}^{\lambda}$ satisfies $\psi''(\lambda) = -\mathrm{e}^{\lambda} < 0$, so ψ is concave; since $\psi(0) = 0$ and $\psi(1) = 3 - \mathrm{e} > 0$, concavity puts ψ above the chord joining these values, hence $\psi \geq 0$ on $[0, 1]$.

Substituting A and B into (13),

$$R'(u) \geq \frac{2\lambda}{1 + u} - \frac{\mathrm{e}^{\lambda} - 1}{1 + u} = \frac{2\lambda - \big(\mathrm{e}^{\lambda} - 1\big)}{1 + u} \geq 0.$$

Therefore R is non-decreasing with $R(0) = 0$, so $R \geq 0$, so $F' \geq 0$ on $[1, \infty)$, and with $F(1) = 0$,

$$\boxed{F(x) \geq 0 \qquad (x \geq 1).}$$

6. Conclusion

For $0 < \lambda \le 1$, that is $1 < a \le \mathrm{e}$, Sections 4 and 5 together with (3) give

$$F(0) = 0, \qquad F \ge 0 \text{ on } (0,1), \qquad F(1) = 0, \qquad F \ge 0 \text{ on } [1,\infty),$$

so the inequality holds on all of $[0,\infty)$. Section 3 shows it fails for $a > \mathrm{e}$. Hence

$$\boxed{(x-1)a^x + 1 \ge x^a \quad \text{for all } x \ge 0 \iff 1 < a \le \mathrm{e}.}$$

7. Numerical picture of the threshold

Values of $F(x) = 1 + (x-1)a^x - x^a$:

x	$a = 2$	$a = \mathrm{e}$	$a = 3$
0.01	0.003014	0.0000467	−0.000937
0.05	0.013998	0.001002	−0.003769
0.10	0.025404	0.003433	−0.005511
0.20	0.041041	0.010288	−0.004585
0.50	0.042893	0.023684	0.008975
1.00	0	0	0
2.00	1.000000	1.808170	2.000000

For $a = 2 < \mathrm{e}$ the values are comfortably positive; at $a = \mathrm{e}$ they are positive but tiny near the origin, consistent with $F \sim \frac{x^2}{2}$ from Section 3; for $a = 3 > \mathrm{e}$ a genuine dip appears immediately to the right of 0 and closes only near $x \approx 0.4$, matching $F(x) \sim (1 - \log 3)x$ with $1 - \log 3 = -0.0986$.

8. Remarks on the structure

The computations are routine; the choice of structure is not.

Where the threshold lives. It is determined entirely at $x = 0^+$, by the linear coefficient $1 - \log a$ in (4). Everything after Section 3 is the verification that this necessary condition is also sufficient, which is not automatic: F has a second zero at $x = 1$ where it could in principle cross.

Why cut at $x = 1$. Because that is the second contact point. Equivalently, the inequality $1+(x-1)a^x \ge x^a$ can be rewritten, for $x \ne 1$, as a comparison involving

$$\frac{x^a - 1}{x - 1},$$

the slope of the secant of $t \mapsto t^a$ through $t = 1$. The substitutions $y = 1 - x$ in Section 4 and $u = x - 1$ in Section 5 both recentre at that point, which is why each side yields a function vanishing at the origin with a sign-definite derivative.

Two different mechanisms on the two sides. On $(0,1)$ the direct approach through F' is awkward, because F starts and ends at 0 and so F' must change sign; instead one compares logarithmic derivatives, which is the right tool for showing that one function dominates a pure power. On $[1,\infty)$ no such subtlety arises: F starts at 0 and is shown to be non-decreasing outright. Attempting a single uniform argument on $[0,\infty)$ is possible but produces a quotient of exponentials with a quadratic numerator, and hides the role of the double contact.